COLLEGE OF ENGINEERING & TECHNOLOGY

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Course Title	: Computer Networks
Course Code	: CC431
Academic Year	: 2026 Summer

Final Cisco Packet Tracer Project Requirements

Objective

Design and implement a complete network using Cisco Packet Tracer that demonstrates the networking concepts covered during the semester.

1. Network Topology

- Create a network containing at least:
- Routers
- Switches
- PCs
- Server

2. IP Addressing

- Use proper IPv4 addressing.
- Divide the network into at least two different subnets.
- Configure the Default Gateway for all devices.

3. Router Configuration

- Configure interface IP addresses.
- Configure Static Routing.
- Save the router configuration.
- At least on one router:
  - Configure Enable Secret Password.
  - Configure Console Password.
  - Configure a Banner Message.

4. Switch Configuration

- Configure Hostname.
- Configure Enable Secret Password.
- Configure Console Password.
- Configure a Banner Message.

5. Server Services

- Configure THREE of the following services:
- DNS

- HTTP
- Email
- Demonstrate that all configured services are functioning correctly.

6. Connectivity Testing

- Successful Ping.
- Successful Traceroute.
- Access the web page using the DNS name.
- FTP connection (if configured).

7. Network Documentation

- Include Network Topology.
- IP Addressing Table.
- Explanation of each configured service.
- Screenshots of testing results.

8. Creativity

- Choose a realistic scenario (Company, Hospital, School, Bank, University, Hotel, etc.).
- Well-organized topology with proper device naming.

Bonus (+2 Marks, Optional)

- Connect multiple networks correctly.
- Use ACLs (if covered).
- Configure DHCP on the router instead of the server.
- Professional topology design and documentation.